Task 1: Company Management System – Data Insertion

You are working with a company database that includes employees, departments, and projects.

Tables:
Employees(EmployeeID, Name, DepartmentID, Salary)
Departments(DepartmentID, DepartmentName)
Projects(ProjectID, ProjectName, DepartmentID)

Tasks:

Write SQL INSERT INTO statements to:

- Add at least 5 departments
- Add at least 10 employees linked to departments
- Add at least 5 projects linked to departments
- Ensure some departments have multiple employees
- Ensure correct matching between foreign keys

Requirements:

- Use INSERT INTO properly
- Maintain correct relationships between tables
- Use realistic company data

Task 2: Hospital Management System – Data Insertion

You are working with a hospital database that includes patients, doctors, and appointments.

Tables:
Patients(PatientID, Name, Age, Gender)
Doctors(DoctorID, Name, Specialty)
Appointments(AppointmentID, PatientID, DoctorID, Date)

Tasks:

Write SQL INSERT INTO statements to:

- Add at least 6 patients

- Add at least 4 doctors with different specialties
- Add at least 8 appointments linking patients and doctors
- Ensure some patients have multiple appointments
- Ensure correct matching between foreign keys

Requirements:

- Use INSERT INTO properly
- Maintain correct relationships between tables
- Use realistic hospital data

Task 3: Company Management System – Data Update

You are working with a company database that includes employees, departments, and projects.

Tables:
Employees(EmployeeID, Name, DepartmentID, Salary)
Departments(DepartmentID, DepartmentName)
Projects(ProjectID, ProjectName, DepartmentID)

Tasks:

Write SQL UPDATE statements to:

- Increase salary of employees in a specific department by 10%
- Change the department name for one department
- Update the project name for one project
- Assign an employee to a different department
- Correct a wrong salary value for a specific employee

Requirements:

- Use UPDATE statement correctly
- Use WHERE clause to target specific rows
- Avoid updating all rows accidentally

- Maintain data consistency

Task 4: Hospital Management System – Data Update

You are working with a hospital database that includes patients, doctors, and appointments.

Tables:
Patients(PatientID, Name, Age, Gender)
Doctors(DoctorID, Name, Specialty)
Appointments(AppointmentID, PatientID, DoctorID, Date)

Tasks:

Write SQL UPDATE statements to:

- Update patient age after correction
- Change doctor specialty for one doctor
- Update appointment date for a specific appointment
- Correct patient name spelling mistake
- Reassign an appointment to a different doctor

Requirements:

- Use UPDATE properly
- Always use WHERE condition
- Update only required records
- Ensure relationships remain valid

Task 5: Company Management System – Data Deletion

You are working with a company database that includes employees, departments, and projects.

Tables:
Employees(EmployeeID, Name, DepartmentID, Salary)
Departments(DepartmentID, DepartmentName)
Projects(ProjectID, ProjectName, DepartmentID)

Tasks:

Write SQL DELETE statements to:

- Delete an employee by EmployeeID
- Remove employees from a specific department
- Delete a project that is no longer active
- Remove a department after reassigning or deleting related data
- Delete employees with salary below a specific value

Requirements:

- Use DELETE statement correctly
- Always use WHERE clause
- Avoid deleting all records accidentally
- Maintain database consistency

Task 6: Hospital Management System – Data Deletion

You are working with a hospital database that includes patients, doctors, and appointments.

Tables:
Patients(PatientID, Name, Age, Gender)
Doctors(DoctorID, Name, Specialty)
Appointments(AppointmentID, PatientID, DoctorID, Date)

Tasks:

Write SQL DELETE statements to:

- Delete a patient record by PatientID
- Remove a doctor who is no longer working
- Delete an appointment after it is completed or canceled
- Remove patients with invalid or missing data
- Delete all appointments for a specific doctor

Requirements:

- Use DELETE properly

- Always include WHERE condition
- Ensure related data is handled carefully
- Avoid accidental full-table deletion

Task 7: Company Management System – Table Removal

You are working with a company database that includes employees, departments, and projects.

Tables:
Employees(EmployeeID, Name, DepartmentID, Salary)
Departments(DepartmentID, DepartmentName)
Projects(ProjectID, ProjectName, DepartmentID)

Tasks:

Write SQL DROP TABLE statements to:

- Drop a table that is no longer required in the system
- Remove a backup or temporary table
- Drop a project-related table after system redesign
- Identify which tables should NOT be dropped due to dependencies
- Practice safe deletion planning before execution

Requirements:

- Use DROP TABLE correctly
- Understand table dependencies
- Avoid dropping required tables
- Recognize impact on relationships

Task 8: Hospital Management System – Table Removal

You are working with a hospital database that includes patients, doctors, and appointments.

Tables:
Patients(PatientID, Name, Age, Gender)
Doctors(DoctorID, Name, Specialty)
Appointments(AppointmentID, PatientID, DoctorID, Date)

Tasks:

Write SQL DROP TABLE statements to:

- Drop a table that is no longer used in the system
- Remove a temporary reporting table
- Simulate system redesign by dropping one module table
- Identify tables that should be dropped last due to dependencies
- Understand impact on patient and appointment data

Requirements:

- Use DROP TABLE correctly
- Be aware of foreign key relationships
- Avoid accidental loss of critical data
- Understand permanent deletion of structure